

Duncan Boyd (he/him) M.Sc. in Electrical Engineering Student

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Technical Skills

- **Programming:** C, Python, MATLAB, HTML
- **Design:** SolidWorks, EasyEDA, Onshape, Minitab
- **Hardware:** Vector Network Analyzer, Spectrum Analyzer, Soldering
- **Documentation:** MS Office, Git, LaTeX

Education

University of British Columbia

Graduated April 2025

BASc., Electrical engineering

University of Calgary

Expected Graduation August 2027

M.Sc., Electrical engineering with medical imaging specialization

Work Experience

IKOMED Technologies Inc., Vancouver, BC

January 2024 – August 2024

Firmware/Software Engineering Intern

- Analyzed electromagnetic simulation data and test data using Python scripts for data-driven design of a microwave ablation medical device.
- Developed resilient C firmware to manage data collection using ten BLE sensors with resistance to connection dropout.
- Built a Python GUI and APIs for multiple USB devices with a user-friendly design and fault-tolerant protocols to handle failures.

Cellcentric GmbH & Co. KG, Burnaby, BC

May 2023 – December 2023

Manufacturing Engineering Intern

- Monitored hydrogen fuel cells for leaks and created a streamlined Excel sheet to find failure patterns.
- Redesigned existing quality tests for prototype cells using ISO standards and confirmed their effectiveness by collecting and analyzing test data.
- Performed statistical analysis using Minitab and compiled it in presentations to inform production decisions.

University of Calgary, Calgary, Alberta

May 2022 – August 2022

Student Researcher

- Improved a machine learning approach to MRI image reconstruction resulting in 20% fewer training cycles.
- Used Tensorflow/Keras to evaluate and tune models on a cluster computing service.
- Summarized research with abstracts, posters, and presentations at two symposiums.

Student Teams

UBC Red Cross Club, University of British Columbia

May 2023 – April 2025

Treasurer, Marketing Director

- Organized finances using Excel sheets and a detailed file structure to improve transparency for over 200 members.
- Lead a team of students to produce posters and social media content for committee events.
- Developed a website for the club with extensive documentation for future directors.

UBC Aerodesign Team, University of British Columbia

September 2021 – April 2024

Advanced Fuselage Member, Advanced Sensors and Controls Member

- Designed remote control aircraft landing gear using SolidWorks and evaluated design using finite element analysis.
- Tested, calibrated, and integrated an airspeed sensor onto a flight control board using Altium.
- Integrated co-dependent engineering components with a 70 member team.

Engineers Without Borders UBC, University of British Columbia

September 2022 – August 2023

Youth Venture Member

- Created and ran a design competition and CAD workshop with the goal of introducing high school students to engineering.
- Learned Onshape (CAD) and video editing software to create instructional videos for over 100 students.
- Answered questions and provided guidance for students while they learned basic engineering design and CAD skills.

Technical Projects

Dashlight, Car Dashboard Display

October 2023 – December 2024

- Build a PIC32-based 7-segment display to read live data from my car's onboard computer.
- Debugged real-time communication issues over a UART connection.

Billboard, Personal Website

April 2025 – May 2025

- Wrote a website using raw HTML/CSS/JS to showcase my experiences.
- Explored dynamic elements and validated design for various screen sizes and devices.

Foghorn, FM Walkie Talkie

May 2025

- Constructed an ESP32-based handheld FM transmitter for sending voice and audio input to car radios.
- Managed I2C peripherals and designed for a simple and safe user experience.

Awards and Certifications

- Alberta Graduate Excellence Scholarship - \$11,000 (2025)
- Technological Stewardship Practice Program (2024)
- Certified SolidWorks Associate (2021)
- DELF B2 French Certification (2020)

Interests and Hobbies

- Alpine skiing, biking, hiking, and paddling
- Building electronics projects
- 3D modelling and 3D printing
- Volunteering in my community